

Session 4 — Analysis II

Exercise sheet

Functions with several inputs, functions with several outputs, and what happens when you write a differential equation for each of them at once. Two examples recur: the squared-error surface of a fitted model, whose bottom is found by following the gradient, and a two-accumulator decision model, whose behaviour is decided by the eigenvalues of a 2×2 matrix. Show your reasoning; most answers are short.

Every exercise opens with a shaded box naming the one rule or definition it rests on, so that the sheet can be attempted by someone who missed the lecture. The box gives you the tool, never the answer.

Three notations, and the name of each.

$f'(t)$	the derivative of f	a <i>number</i> : the slope of $f : \mathbb{R} \rightarrow \mathbb{R}$ at t
$\nabla f = (\partial_x f, \partial_y f)$	the gradient of f	a <i>vector</i> , for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$
$\mathbf{J} = \begin{pmatrix} \partial_x F_1 & \partial_y F_1 \\ \partial_x F_2 & \partial_y F_2 \end{pmatrix}$	the Jacobian of $\mathbf{F}$	a <i>matrix</i> , for $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

A dot on top, as in $\dot{x}$, always means d/dt .

Exercise 1 Partial derivatives. Compute $\partial f / \partial x$ and $\partial f / \partial y$.

What this exercise uses. A **partial derivative** is an ordinary derivative taken with the other variable *frozen*: to compute $\partial f / \partial x$, treat y as a constant and differentiate in x with the one-variable rules of Session 3 — the power rule, the rule $(e^{at})' = a e^{at}$, and the chain rule. Nothing new is being defined: the symbol ∂ instead of d is only a reminder that something has been held fixed.

- (a) $f(x, y) = x^2 + 3xy + y$;
- (b) $f(x, y) = x^2 + 4y^2$;
- (c) $f(x, y) = x^2 y + \sin y$;
- (d) $f(x, y) = e^{-(x^2 + y^2)}$.

Exercise 2

What this exercise uses. A partial derivative is a *function of both variables*, not a number: to evaluate it you substitute both coordinates of the point. Geometrically it is the slope of the surface in the x direction, *measured at a specific point* — and the surface may tilt differently in the x direction depending on how far along y you are standing.

- (a) For $f(x, y) = x^2 + 3xy + y$, evaluate $\partial f / \partial x$ at $(1, 0)$ and at $(1, 2)$.
- (b) In one sentence: why are the two answers different, and what does that tell you about functions of several variables?

Exercise 3 Contours and the gradient.

What this exercise uses. A **contour** (level set) of f is a curve $\{f = c\}$: all the points at the same height. The **gradient** $\nabla f = (\partial_x f, \partial_y f)$ carries two pieces of information at once — its *direction* is the direction of steepest increase, and its *length* $\|\nabla f\|$ is the rate along that direction. It is perpendicular to the contour through the point, because a curve $\mathbf{c}(s)$ running along a contour keeps f constant, so $0 = \frac{d}{ds}f(\mathbf{c}(s)) = \nabla f \cdot \mathbf{c}'(s)$. Along a unit direction $\mathbf{u}$ the rate of change is $\nabla f \cdot \mathbf{u}$.

- (a) Sketch the contour lines $f = 1$ and $f = 4$ of $f(x, y) = x^2 + 4y^2$.
- (b) Compute $\nabla f(1, 1)$.
- (c) The contour through $(1, 1)$ has tangent direction $(-8, 2)$. Verify that $\nabla f(1, 1)$ is orthogonal to it.
- (d) In which direction from $(1, 1)$ does f increase fastest, and at what rate?

Exercise 4 An iso-performance curve — all the combinations of task difficulty d and practice p giving the same accuracy — is a contour line of the accuracy function.

Take $A(d, p) = 1 - e^{-p/(2d)}$ with $d > 0$, $p \geq 0$.

What this exercise uses. Two ideas meet here. First, partial derivatives are read as *signs*: a positive partial means “increasing this input increases the output, *the other input held fixed*”. Second, finding an iso-curve means solving $A(d, p) = c$ for one variable in terms of the other, and the tool for undoing an exponential is the logarithm: $e^u = v \iff u = \ln v$ (for $v > 0$).

- (a) Compute $\partial A / \partial p$. Is accuracy increasing or decreasing in practice?
- (b) Compute $\partial A / \partial d$ and give its sign. Interpret it in one sentence.
- (c) Show that the contour $A = c$ is a straight line through the origin in the (d, p) plane, and give its slope in terms of c .

Exercise 5 Fitting a model means finding the bottom of an error surface. With one parameter it is a curve; with two it is a bowl over the (w_0, w_1) plane.

Let $J(w) = \sum_{i=1}^n (y_i - wx_i)^2$, a function of the single parameter w .

What this exercise uses. In an **error surface** the parameters are the variables and the data x_i, y_i are *constants* — differentiate with respect to w and nothing else. Two ways to find the bottom: solve $dJ/dw = 0$ exactly, or walk downhill by **gradient descent**, repeatedly stepping along $-dJ/dw$. And an upward parabola $aw^2 + bw + c$ with $a > 0$ has exactly one stationary point, which is its minimum.

- (a) Compute dJ/dw .

- (b) Set it to zero and solve for w . (You have just derived least-squares regression through the origin.)
- (c) Write down one step of gradient descent, $w \leftarrow w - \eta dJ/dw$, and say in words what η controls.
- (d) Why does this J have exactly one critical point, and why is it a minimum?

Exercise 6 Critical points.

What this exercise uses. A **critical point** is a point where $\nabla f = \mathbf{0}$: the surface is locally flat. As in one variable this is *necessary* for an interior maximum or minimum and not sufficient — and in two variables there is a genuinely new third possibility, a point that is a minimum along one direction and a maximum along another. You can classify without any second-derivative machinery by *restricting f to lines through the point* and comparing what happens along each.

- (a) Find all points where $\nabla f = \mathbf{0}$ for $f(x, y) = x^2 + y^2$, and say what kind of point it is.
- (b) Do the same for $f(x, y) = x^2 - y^2$. Describe the surface in words.
- (c) Do the same for $f(x, y) = e^{-(x^2+y^2)}$.

Exercise 7 Trajectories. For each, compute $\dot{\mathbf{x}}(t)$, the speed $\|\dot{\mathbf{x}}(t)\|$, and describe the path.

What this exercise uses. A **vector-valued function of one variable** $\mathbf{x} : \mathbb{R} \rightarrow \mathbb{R}^2$ is differentiated *component by component*. Two different objects come out of it and must not be confused: the **velocity** $\dot{\mathbf{x}}(t)$, a vector giving direction and rate, and the **speed** $\|\dot{\mathbf{x}}(t)\| = \sqrt{\dot{x}_1^2 + \dot{x}_2^2}$, a number. The velocity is always tangent to the path. And the path (the set of points visited) is one thing, the motion along it is another: the same curve can be traced at any speed.

- (a) $\mathbf{x}(t) = (2t, 3 - t)$;
- (b) $\mathbf{x}(t) = (t, t^2)$;
- (c) $\mathbf{x}(t) = (\cos t, \sin t)$;
- (d) $\mathbf{x}(t) = (e^{-t} \cos t, e^{-t} \sin t)$.

Exercise 8 For $\mathbf{y}(t) = (\cos t, \sin t)$:

What this exercise uses. The **dot product** $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2$ is zero exactly when the two vectors are perpendicular. One identity does all the work here:

$$\frac{d}{dt} \|\mathbf{y}\|^2 = \frac{d}{dt} (\mathbf{y} \cdot \mathbf{y}) = 2 \mathbf{y} \cdot \dot{\mathbf{y}},$$

so “ $\mathbf{y} \cdot \dot{\mathbf{y}} = 0$ ” and “ $\|\mathbf{y}\|$ is constant” are the same statement.

- (a) Show that $\mathbf{y} \cdot \dot{\mathbf{y}} = 0$ for every t .
- (b) Explain in one sentence what that says geometrically.
- (c) Give the velocity at $t = 0$ and at $t = \pi/2$, and mark both on a sketch.

Exercise 9 In a mouse-tracking experiment the cursor position $\mathbf{x}(t) = (x(t), y(t))$ is recorded as the participant moves toward one of two response buttons. A path that bends toward the unchosen option is evidence that the two responses competed.

A recorded trajectory is modelled by $\mathbf{x}(t) = (t, t^2(1-t))$ for $t \in [0, 1]$, with the two buttons at $(1, 0)$ and $(1, 1)$.

What this exercise uses. Two things from Session 3, applied to one component of a trajectory. First, $y(t)$ is largest where $\dot{y} = 0$ and $\dot{y}$ changes sign from $+$ to $-$ — but on a *closed* interval the two endpoints must be compared as well, since a maximum can sit at an end without the derivative vanishing there. Second, the velocity vector points along the direction of travel, so its components say where the cursor is heading at that instant. Expand before differentiating.

- (a) Compute $\dot{\mathbf{x}}(t)$.
- (b) Give the velocity at $t = 0$ and at $t = 1$.
- (c) At which t is $y(t)$ largest? What does that instant represent?

Exercise 10 Jacobians. Write $\mathbf{J}(x, y)$ for each, and evaluate it where asked.

What this exercise uses. The **Jacobian matrix** of $\mathbf{F} = (F_1, F_2)$ collects every partial derivative, one *output* per row and one *input* per column:

$$\mathbf{J}(x, y) = \begin{pmatrix} \partial F_1 / \partial x & \partial F_1 / \partial y \\ \partial F_2 / \partial x & \partial F_2 / \partial y \end{pmatrix}.$$

Row i is the gradient of the i -th component. For $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ the Jacobian is $m \times n$ — as many rows as outputs, as many columns as inputs — which is what makes $\mathbf{J}\delta$ a legal product.

- (a) $\mathbf{F}(x, y) = (-x + y, 2x - 3y)$. Why is $\mathbf{J}$ constant?
- (b) $\mathbf{F}(x, y) = (x^2y, x + \sin y)$, at $(1, 0)$.
- (c) $\mathbf{F}(x, y) = (x - x^3 - y, x - y)$, at $(0, 0)$ and at $(1, 1)$.
- (d) For a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, what is the Jacobian, and what is it usually called?

Exercise 11

What this exercise uses. Linearisation. Near a point, a differentiable map is well approximated by an affine one: the value at the point, plus a *linear* map applied to the displacement. That linear map is the derivative, in whatever shape the dimensions require — a number, a row, or a matrix. The error is of order $\|\delta\|^2$, which is why the approximation is claimed only *near* $\mathbf{x}_0$.

- (a) State in one line what $\mathbf{J}$ approximates: $\mathbf{F}(\mathbf{x}_0 + \delta) \approx ?$
- (b) Write the one-variable statement it generalises.
- (c) Use $\mathbf{J}$ to estimate $\mathbf{F}(0.1, -0.1)$ for $\mathbf{F}(x, y) = (x - x^3 - y, x - y)$, given $\mathbf{F}(0, 0) = \mathbf{0}$.

Exercise 12 For each matrix give $T = \text{tr } \mathbf{A}$, $D = \det \mathbf{A}$, the eigenvalues, and the type of flow of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ (stable/unstable node, saddle, spiral, centre).

What this exercise uses. For a 2×2 matrix the characteristic polynomial is $\lambda^2 - T\lambda + D$ with $T = \text{tr } \mathbf{A} = a_{11} + a_{22}$ and $D = \det \mathbf{A} = a_{11}a_{22} - a_{12}a_{21}$, so

$$\lambda_{\pm} = \frac{T \pm \sqrt{T^2 - 4D}}{2}, \quad \lambda_+ + \lambda_- = T, \quad \lambda_+ \lambda_- = D.$$

$D < 0$	saddle (always unstable)
$D > 0, T^2 \geq 4D$	node — stable if $T < 0$, unstable if $T > 0$
$D > 0, T^2 < 4D$	spiral — stable if $T < 0$, unstable if $T > 0$
$D > 0, T = 0$	centre (closed orbits)

The sign of D decides saddle or not, the discriminant $T^2 - 4D$ decides node or spiral, and the sign of T decides stable or unstable — but only once $D > 0$ is known.

- (a) $\begin{pmatrix} -2 & 0 \\ 0 & -5 \end{pmatrix};$
- (b) $\begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix};$
- (c) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix};$
- (d) $\begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix};$
- (e) $\begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}.$

Exercise 13

What this exercise uses. An **eigenvector** $\mathbf{v}$ of $\mathbf{A}$ is a direction the matrix does not rotate: $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, only stretched by λ . Along such a direction the system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ collapses to the scalar equation $\dot{s} = \lambda s$ of Session 3, so a general solution is a superposition of modes, $\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2$. For a complex pair $\lambda = \mu \pm i\omega$, the real part μ sets growth or decay (with $\tau = 1/|\mu|$) and the imaginary part ω sets the rotation, at angular speed ω and period $2\pi/\omega$.

- (a) Verify that $\mathbf{x}(t) = e^{\lambda t} \mathbf{v}$ solves $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ whenever $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$.
- (b) For $\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & -4 \end{pmatrix}$ and $\mathbf{x}(0) = (2, 1)$, write $\mathbf{x}(t)$ explicitly. Which component dies first?
- (c) A system has eigenvalues $-0.1 \pm 3i$. Is it stable? Does it oscillate? At what angular speed, and with what period?
- (d) A system has eigenvalues -100 and $+0.01$. Is it stable? Explain why “mostly negative” is not enough.

Exercise 14 A standard model of two-alternative choice: two evidence accumulators, each decaying and each inhibiting the other with strength $c > 0$.

$$\dot{x} = -x - cy, \quad \dot{y} = -cx - y, \quad \mathbf{A} = \begin{pmatrix} -1 & -c \\ -c & -1 \end{pmatrix}.$$

What this exercise uses. When a matrix has **equal diagonal entries and equal off-diagonal entries**, $\begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}$, its eigenvectors are always $(1, 1)$ and $(1, -1)$, whatever α and β : the matrix treats the two coordinates identically, so it cannot distinguish them. Multiply each of the two out to read off the eigenvalue that goes with it. Once the eigen-directions are known, *decomposing the initial condition along them* turns the system into two independent scalar equations.

- (a) $\mathbf{A}$ is symmetric. Give its two eigenvectors without computing anything, and then its two eigenvalues.
- (b) Interpret each eigenvector in words.
- (c) For which c does the difference $x - y$ grow? What does the model then do?
- (d) Take $c = 2$ and $\mathbf{x}(0) = (1.1, 0.9)$. Write $\mathbf{x}(t)$ and describe its behaviour.
- (e) Compute $D = \det \mathbf{A}$ for $c = 2$ and confirm the flow type.

Exercise 15 Nonlinear systems: nullclines, fixed points, and the Jacobian.

What this exercise uses. For a nonlinear system $\dot{x} = F(x)$:

- a **nullcline** is a curve on which one component of the velocity vanishes — the x -nullcline is $\{F_1 = 0\}$, crossed vertically; the y -nullcline is $\{F_2 = 0\}$, crossed horizontally;
- a **fixed point** is where *both* vanish, i.e. an intersection of the two nullclines — so drawing the nullclines is how you *find* the fixed points;
- near a fixed point x^* , writing $x = x^* + \delta$ gives $\dot{\delta} \approx J(x^*)\delta$, and the classification of that linear system transfers to the nonlinear one **provided no eigenvalue has zero real part**. When one does, the linearisation is silent and the discarded terms decide.

- (a) For $\dot{x} = y, \dot{y} = -x - y$: find the fixed point, write J , give T and D , and classify.
- (b) For $\dot{x} = y - x, \dot{y} = x^2 - y$: find the nullclines, the fixed points, and classify each one.
- (c) For $\dot{x} = x - x^3 - y, \dot{y} = x - y$: find the fixed point and show that the linearisation is inconclusive. What would you do next?

Exercise 16 Putting it together. A two-variable model has a single fixed point, and its Jacobian there is $J = \begin{pmatrix} -1 & \alpha \\ -1 & -1 \end{pmatrix}$, with α a parameter.

What this exercise uses. Everything is read off the table of Exercise 12, now with T and D depending on a parameter. The three questions are three inequalities:

$$\text{stable} \iff T < 0 \text{ and } D > 0, \quad \text{spiral} \iff T^2 < 4D, \quad \text{saddle} \iff D < 0.$$

Note that T here does not depend on α at all, so the whole family lives on a single *vertical line* of the trace-determinant plane and varying α slides the point up and down it. Do not forget to say what happens at the two values of α where the point crosses a boundary.

- (a) Give T and D in terms of α .
- (b) For which α is the fixed point stable?
- (c) For which α is it a spiral rather than a node?
- (d) For which α is it a saddle?